Supporting Information for: Plant-wide Bayesian fault diagnosis in recycle processes via simulation-based inference under multi-loop feedback control

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S1. STOCHASTIC DATA GENERATION AND SCENARIO OVERVIEW

a. Simulation pipeline. Observation windows are generated by integrating the degraded closed-loop model (Eqs. (3)–(6) of the main text) with the Euler–Maruyama scheme at a fixed step of $\Delta t = 0.1$ min. Two independent noise layers are applied. *Process noise* is injected at each integration step as an additive Gaussian increment: $\boldsymbol{\sigma} = [\sigma_C, \sigma_T, \sigma_{T_c}] = [5 \times 10^{-4}$ mol L^{-1} min$^{-1/2}$, 0.1 K min$^{-1/2}$, 0.1 K min$^{-1/2}]$, consistent with mild process variability at steady state. *Sensor noise* is applied post-simulation as a zero-mean Gaussian draw with standard deviation equal to 0.5% of each channel's running maximum, representing a typical industrial-grade sensor specification.

Each replicate is initialised from the nominal closed-loop steady state, computed by integrating the deterministic ODE to convergence with a high-accuracy adaptive solver. Initialising from the steady state removes startup transients and ensures that the 60-minute window is representative of stationary operation around the prescribed (α^*, β^*) operating point. The four observable channels $[C(t), T(t), T_c(t), Q_c(t)]$ are subsampled to a grid of $\Delta t_{\text{out}} = 0.5$ min, yielding one 120×4 observation matrix per replicate.

b. Dataset structure. The full System I dataset comprises 8 scenarios $\times$ 50 independent stochastic replicates $= 400$ observation windows. The eight scenarios (Table 1 of the main text) span open- and closed-loop configurations and a range of fault severities; their ground-truth parameters (α^*, β^*) are summarised in Table 1 of the main text. Each replicate within a scenario shares the same ground-truth (α^*, β^*) but was integrated with an independent random seed, so the 50 replicates per scenario are exchangeable draws from $p(\mathbf{x} \mid \alpha^*, \beta^*)$ at that operating point.

c. Scenario trajectories. Figure 1 shows six representative replicates per closed-loop scenario across all four observable channels. The bold black trace is the per-scenario replicate mean.

Several features are immediately apparent. First, the reactor temperature T is tightly regulated near the setpoint $T_{\text{sp}} = 312.5$ K in all non-saturating scenarios: the PI controller maintains the same temperature profile regardless of whether the jacket is fouled (Sc 2) or the catalyst is decaying (Sc 3), confirming the closed-loop masking mechanism discussed in Section 2.2 of the main text. Second, the coolant flow Q_c shows qualitatively distinct behaviour across scenarios: jacket fouling (Sc 2) drives Q_c to ≈ 210 L min^{-1}, while catalyst decay (Sc 3) lowers it to ≈ 50 L min^{-1}, and severe fouling (Sc 5) saturates the valve at $Q_{c,\max} = 400$ L min^{-1}. Third, the concentration channel C shifts distinctly between scenarios with different α^*: catalyst decay (Sc 3, $\alpha^* = 0.70$) raises C_{ss} from ≈ 0.018 to ≈ 0.026 mol L^{-1}, while jacket fouling (Sc 2) produces a modest downward shift attributable to the residual thermal coupling through T_c. The sensor-drift scenario (Sc 7, $\delta_T = +2$ K) is indistinguishable from Sc 1 in the C and T_c channels but shows a systematic upward offset in the T channel, motivating the four-dimensional $(\alpha, \beta, \delta_T, \delta_{C_i})$ extension explored in future work.

S2. PRIOR PREDICTIVE CHECK

A necessary condition for valid amortised inference is that the prior-predictive distribution — the distribution of summary statistics produced by prior draws — covers the evaluation observations. To verify this, we drew 500 parameter vectors $\boldsymbol{\theta} = (\alpha, \beta)$ from the uniform prior $\mathcal{U}([0.4, 1.2]^2)$ and simulated each through the stochastic closed-loop model to obtain synthetic 60-minute observation windows. For every feature in the 29-D summary vector, the [2.5%, 97.5%] interval of the prior-predictive distribution was compared against the observed values from all six closed-loop evaluation scenarios (Sc 1–5, Sc 7). At least 80% of the observed values per feature fall inside this interval for the large majority of the 29-D summary vector, confirming that the simulator faithfully spans the evaluation regime. Under the matched-protocol warm-start pipeline used for the production posterior (Section 6.1 of the main text), three temperature-related features (T_{mean}, $T_{\text{final_mean}}$, $\int |T - T_{\text{sp}}| \, \mathrm{d}t$) fall below the 80% threshold when pooled across all six closed-loop scenarios (72.7%, 76.0%, and 70.7% respectively at $N_{\text{prior}} = 500$; stable to within Monte Carlo noise up to $N_{\text{prior}} = 2{,}000$, confirming this is not a sample-size effect — see below).

Root cause, identified by decomposing the pooled coverage per scenario. The shortfall is not spread across the six evaluation scenarios: it is almost entirely attributable

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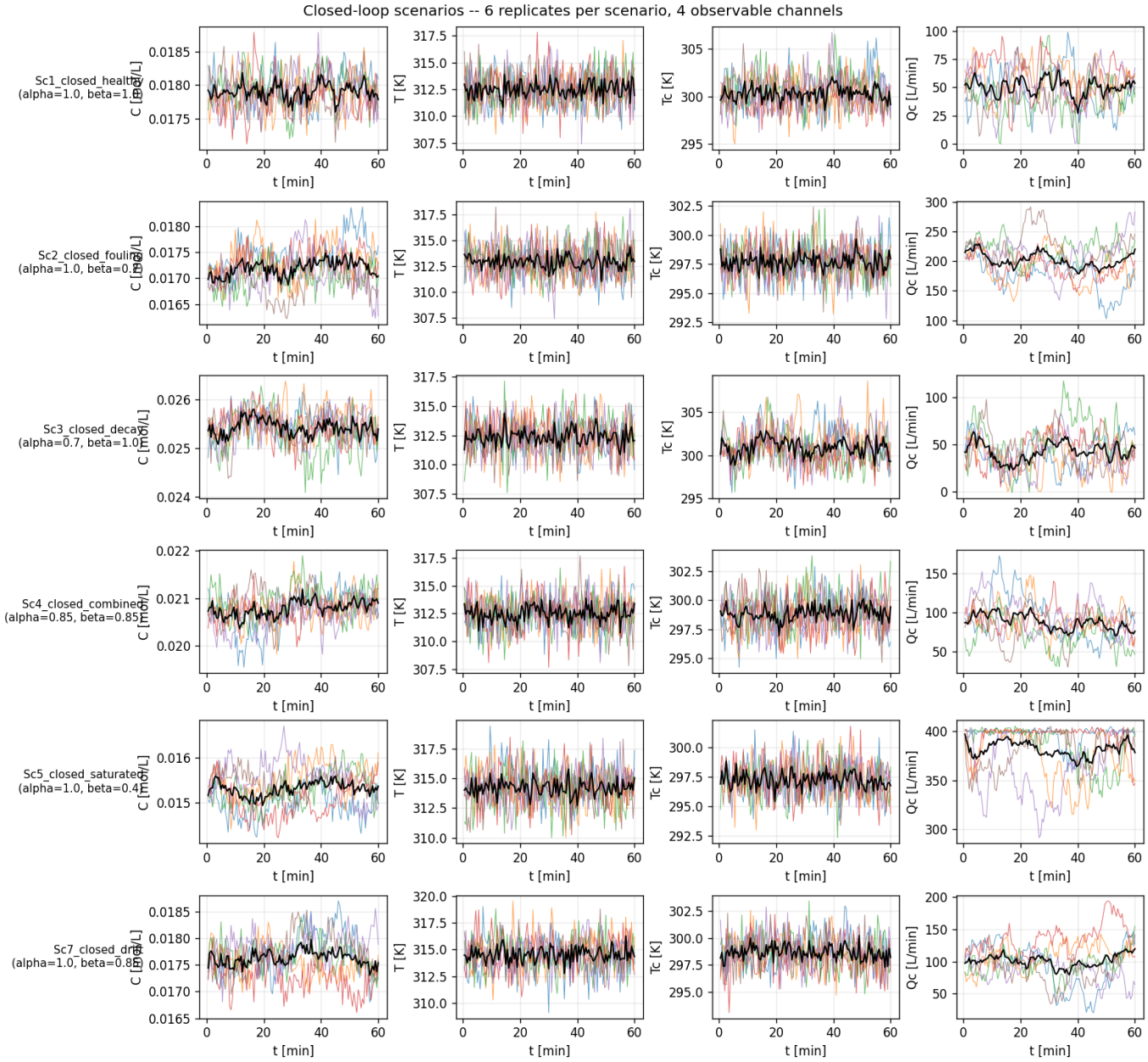


FIG. 1: Representative observation windows for the six closed-loop fault scenarios (Sc 1–5, Sc 7). Each row corresponds to one scenario (scenario name and ground-truth (α^*, β^*) shown on the left); columns show the four observable channels $[C, T, T_c, Q_c]$ over the 60-minute window. Coloured traces: six individual stochastic replicates. Bold black trace: replicate mean. The reactor temperature T (second column) is regulated near $T_{\mathrm{sp}} = 312.5$ K across all scenarios; the fault signal is encoded primarily in Q_c (fourth column) and C (first column), consistent with the Fisher-information analysis in Section 4.3.1 of the main text.

to Sc 7, the sensor-drift scenario. Sc 7's temperature-related feature values lie *entirely outside* the prior-predictive envelope (0%, 12%, and 2% per-scenario coverage for the three features respectively) because its +2 K temperature offset is an additive sensor bias applied after simulation (δ_T, Section S1), not a function of the closed-loop dynamics that the 2-D (α, β) prior parameterises — no draw from $\mathcal{U}([0.4, 1.2]^2)$ can reproduce it, regardless of how many simulations are drawn. Excluding Sc 7, pooled coverage across the remaining five scenarios rises to 87.2%, 88.8%, and 84.4% respectively, comfortably above threshold; the residual imperfection there comes from Sc 5 (severe fouling, actuator saturation), whose coverage is 40–54% per feature and which — unlike Sc 7 — *is* a function of (α, β) and would respond to a wider prior, since a broader box increases the fraction of prior

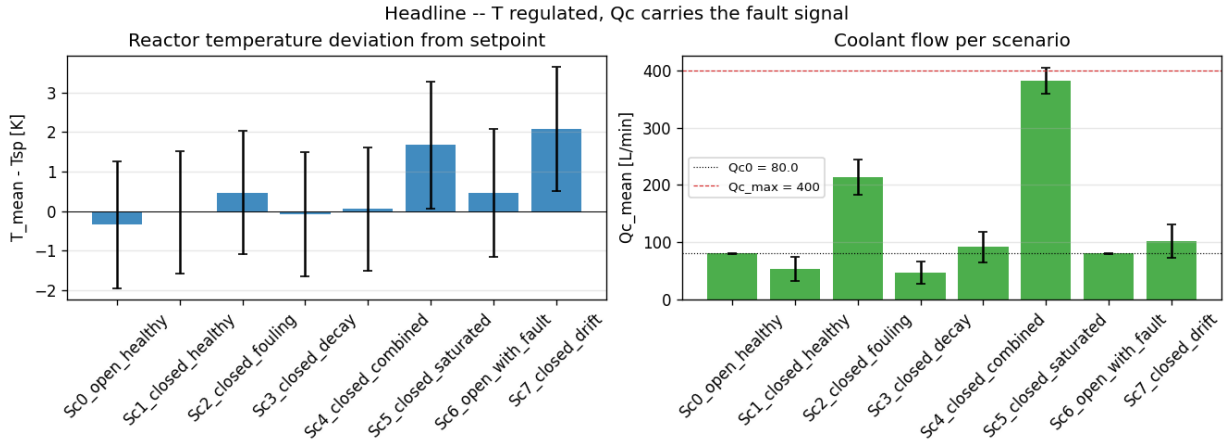


FIG. 2: Headline summary of the closed-loop masking effect across all eight scenarios (50 stochastic replicates each, 60-min windows). *Left:* Mean reactor temperature deviation from setpoint $\bar{T} - T_{\mathrm{sp}}$ with $\pm 1\sigma$ error bars. Under non-saturating PI control, $\bar{T} - T_{\mathrm{sp}} \approx 0$ for all closed-loop scenarios regardless of fault severity. *Right:* Mean coolant flow rate $\bar{Q}_c$ per scenario. Jacket fouling (Sc 2) raises $\bar{Q}_c$ to ≈ 210 L min^{-1}; catalyst decay (Sc 3) reduces it to ≈ 50 L min^{-1}; severe fouling (Sc 5) saturates the valve at $Q_{c,\max} = 400$ L min^{-1}. The fault signal lives entirely in the actuator output, not in the regulated reactor temperature.

draws that saturate the actuator.

We also directly tested whether more prior simulations close this gap: repeating the check at $N_{\mathrm{prior}} \in \{500, 1{,}000, 1{,}500, 2{,}000\}$ from one shared pool of 2,000 draws changes the $[2.5\%, 97.5\%]$ interval width by *less than 3%* and the resulting coverage by less than 1 percentage point at every budget tested, confirming $N_{\mathrm{prior}} = 500$ was already fully converged for this diagnostic — the shortfall is a scope mismatch (Sc 7's δ_T dimension is absent from the prior being checked), not a Monte Carlo sampling-noise artefact, and is not expected to affect any reported posterior-accuracy or classification result, all of which condition on the observed data directly rather than on this prior-predictive coverage statistic.

Figure 3 illustrates the check for four representative features, including the two physics-informed proxies that carry most of the identifying information about (α, β) (see Section 3.1 of the main text). The observed scenario means (vertical markers) are well contained within the prior-predictive histograms for every feature shown, with no scenario lying systematically outside the prior envelope.

S3. TRAINING BUDGET SENSITIVITY

To confirm that the choice of $n_{\mathrm{sim}} = 10{,}000$ training simulations is sufficient, we trained three posteriors using MAF density estimators, under the same matched scenario-specific warm-start protocol as the production posterior (Section 6.1 of the main text), at budgets of 1,000, 5,000, and 10,000 prior draws, and evaluated each on the healthy scenario (Sc 1, $\alpha^* = \beta^* = 1.0$) and the jacket-fouling scenario (Sc 2, $\beta^* = 0.70$).

For Sc 2, the posterior mean of β is $\hat{\beta} = 0.747$ at $n_{\mathrm{sim}} = 1{,}000$, 0.735 at $n_{\mathrm{sim}} = 5{,}000$, and 0.733 at $n_{\mathrm{sim}} = 10{,}000$ — essentially flat beyond $n_{\mathrm{sim}} = 5{,}000$, which is the basis for choosing $n_{\mathrm{sim}} = 10{,}000$ for the production posterior. 90% credible-interval coverage of the true parameters on Sc 1 is achieved at all three budgets for α, and at $n_{\mathrm{sim}} \geq 5{,}000$ for β ($n_{\mathrm{sim}} = 1{,}000$ misses); on Sc 2, β's 90% interval contains the true value at $n_{\mathrm{sim}} = 5{,}000$ but not at 1,000 or 10,000 (a single-replicate diagnostic, not a calibration claim — see Section S6 for the actual SBC-based calibration assessment across many replicates).

Discrepancy with the production posterior, and why it is expected. This MAF ablation's $\hat{\beta} \approx 0.73$–0.75 on Sc 2 *overshoots* the true value of 0.70, whereas the production NSF posterior (Section 6.1, Table VI of the main text) gives $\hat{\beta} = 0.6978$, i.e. essentially unbiased. Both use the matched warm-start protocol, so the discrepancy is not a training/evaluation mismatch; it reflects (i) a different density-estimator family (MAF here vs. the production NSF) and (ii) a different random seed. This is expected variation across the design choices this ablation exists to explore, and does not indicate a problem with the production posterior used everywhere else in the manuscript — Figure 4 below should be read as a budget-sensitivity comparison among these three MAF runs, not as an alternative estimate of the production posterior's accuracy.

I. SUMMARY STATISTICS

a. Group 1 — Per-channel base statistics (20 features). For each of the four observable channels $[C, T, T_c, Q_c]$, five scalar statistics are computed over the full 60-minute window: the arithmetic mean, standard

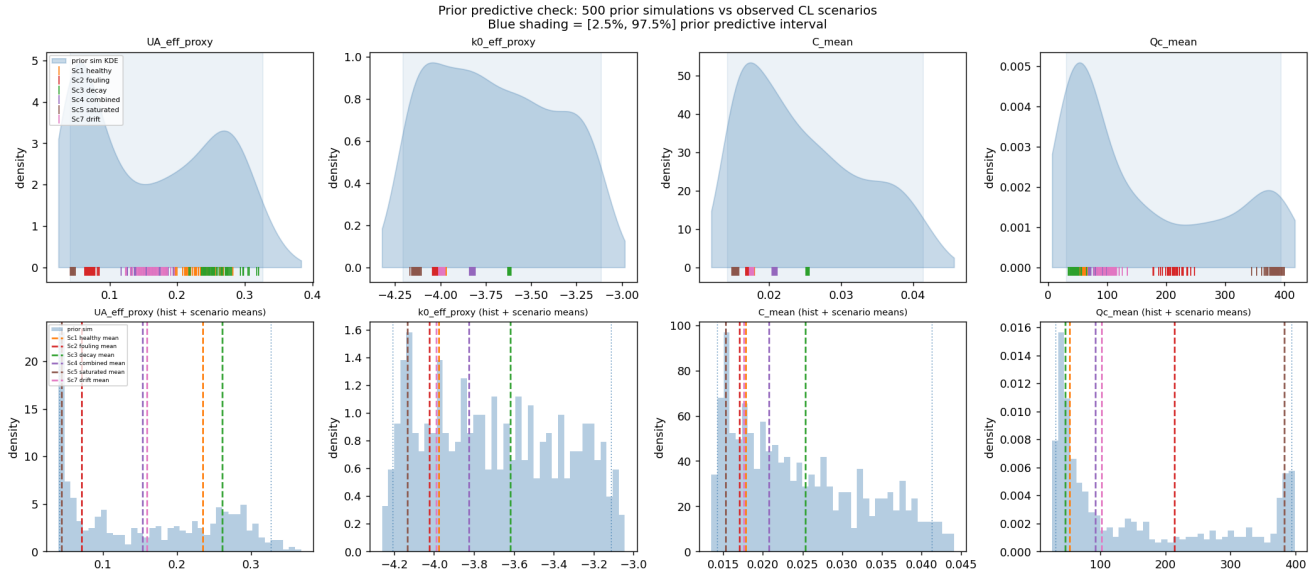


FIG. 3: Prior predictive check ($N = 500$ prior draws) for four key summary features: s_{UA}, s_{k_0}, $\bar{C}$, and $\bar{Q}_c$. *Top row:* kernel density estimates of the prior-predictive distribution (blue fill) with observed values from each closed-loop scenario overlaid as tick marks. *Bottom row:* histogram of prior-predictive values with per-scenario means marked as vertical lines. All scenario observations fall within the prior-predictive envelope, confirming that the simulator covers the operating regime of the evaluation set.

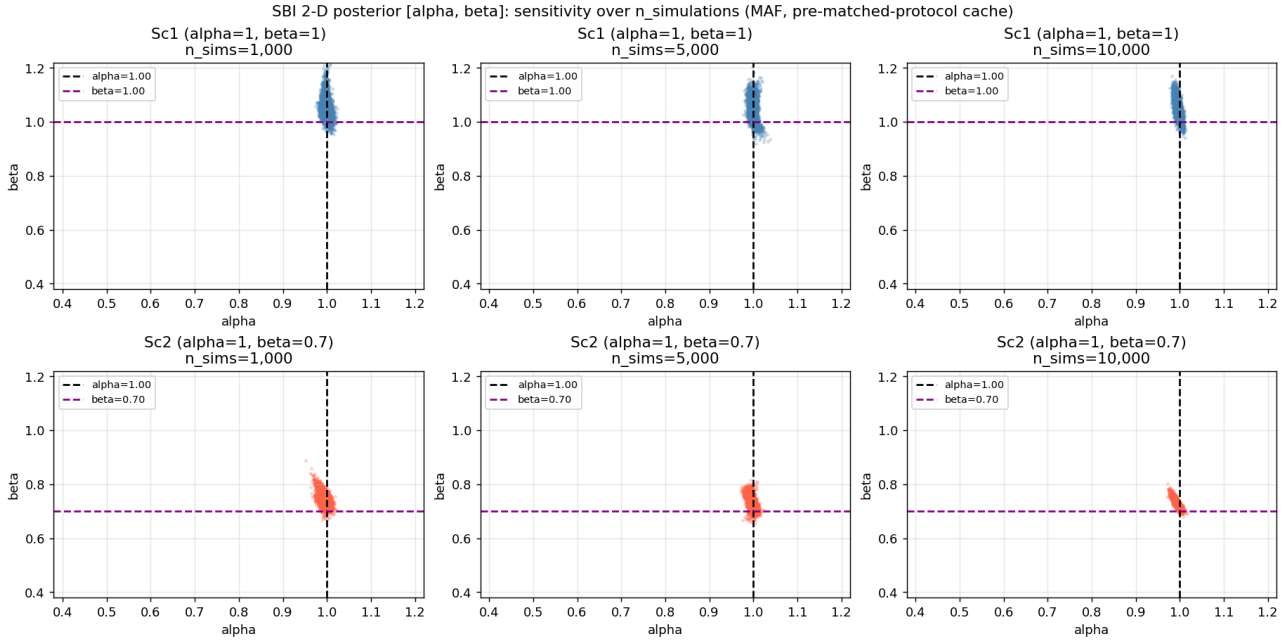


FIG. 4: Training budget sensitivity: 2-D posterior scatter plots $[\alpha, \beta]$ for Sc 1 (healthy, top row) and Sc 2 (jacket fouling, bottom row) at three simulation budgets (MAF estimator, matched scenario-specific warm-start protocol). Dashed lines mark the true parameter values. The Sc 2 β posterior mean ($\hat{\beta} \approx 0.73$–0.75, true 0.70) changes little beyond $n_{\text{sim}} = 5{,}000$, motivating the production budget of 10,000; this MAF ablation's small residual overshoot is a property of this specific estimator/seed combination and is larger than the production NSF posterior's bias ($\hat{\beta} = 0.6978$, Section 6.1 of the main text) — see the discussion text for why the two are not expected to match exactly.

deviation, ordinary-least-squares slope, minimum, and maximum. These 20 features capture the level, variability, trend, and range of each signal and form the baseline subset against which the additional groups are evaluated.

b. Group 2 — Final-window means (4 features). The arithmetic mean of each channel over the final 25% of the observation window (minutes 45–60) provides an estimate of the current quasi-steady-state operating point, less contaminated by fault-onset transients than the whole-window mean. Under stationary closed-loop operation these four values converge to the steady-state measurements C_{ss}, T_{ss}, $T_{c,\mathrm{ss}}$, and $Q_{c,\mathrm{ss}}$.

c. Group 3 — Control aggregates (3 features). Three aggregates quantify the operational burden placed on the PI controller during the observation window:

- $\int_0^{60} |T(t) - T_{\mathrm{sp}}|\,\mathrm{d}t$: the time-integrated absolute temperature error, which increases whenever the controller is unable to maintain the setpoint, most prominently during the actuator-saturation regime of Sc 5.
- $f_{\mathrm{sat,\,low}}$: the fraction of time steps at which $Q_c = Q_{c,\min} = 0$ L min^{-1}, indicating fully-closed valve saturation.
- $f_{\mathrm{sat,\,high}}$: the fraction of time steps at which $Q_c = Q_{c,\max} = 400$ L min^{-1}, indicating fully-open saturation.

All three aggregates equal zero under healthy, non-saturating closed-loop operation and become the dominant diagnostic features in the degraded regimes where the per-channel statistics carry reduced discriminating power.

d. Group 4 — Physics-informed proxies (2 features). Two features are derived by algebraic inversion of the closed-loop steady-state balance equations, making them direct proxies for the identifiable parameter products $\beta \cdot UA$ and $\alpha \cdot k_0$:

$$s_{UA} = \frac{\bar{T} - \bar{T}_c}{\bar{Q}_c}, \tag{1}$$

$$s_{k_0} = \ln\!\left(\frac{\bar{C}}{C_i - \bar{C}}\right), \tag{2}$$

where $\bar{(\cdot)}$ denotes the whole-window mean of the respective channel. Under non-saturating PI control at steady state, $s_{UA} \propto (\beta \cdot UA)^{-1}$ follows from the jacket energy balance, and $s_{k_0} \propto (\alpha \cdot k_0)^{-1}$ follows from the component balance at fixed conversion. Used alone, these two features achieve 99.3% cross-validated Linear Discriminant Analysis (LDA) [1] accuracy on the eight-scenario dataset, confirming that they encode nearly all of the identifiable information about (α, β) accessible from stationary, noise-free steady-state observations. Figure **??** illustrates this separation directly: all eight scenarios occupy well-separated, compact clusters in the $(s_{UA},\, s_{k_0})$ plane.

S3A. EXTENDED HYPERPARAMETER ABLATION STUDY

A reviewer of a closely related paper by the same group (Casolo et al., *Engineering Applications of Artificial Intelligence*, in press) requested that training-set size, density-estimator depth, and optimiser/architecture choices be varied and justified rather than fixed without comment. This section reports that ablation for System I, across three axes: training budget ($n_{\mathrm{sim}} \in \{500, 1{,}000, 2{,}500, 5{,}000, 10{,}000\}$), Neural Spline Flow (NSF) architecture width and depth (hidden features × transforms $\in \{(64, 3), (128, 5)\,[\text{production}], (256, 5), (128, 8)\}$), and density-estimator family (NSF vs. Masked Autoregressive Flow, MAF, at the production width/depth). All nine configurations are trained under the same matched scenario-specific warm-start protocol as the production posterior (Section 6.1 of the main text), on prefixes of one shared 10,000-draw training bank (so the budget axis tests genuinely nested subsets, not independent re-draws), and evaluated against one shared, 300-draw held-out simulation-based-calibration (SBC) set never used for training.

Point accuracy is stable across the entire grid; formal joint calibration is not, and this is a genuine, reportable finding rather than a training bug. The Sc 2 β posterior mean stays within 0.690–0.728 across all nine configurations — a ±5% band around the true value of 0.70 regardless of budget, width, depth, or estimator family, confirming that $n_{\mathrm{sim}} = 10{,}000$ with the production architecture is not an arbitrary choice: smaller budgets and smaller networks are already adequate for point estimation, and larger ones buy nothing more. Formal SBC calibration is a different story: **none of the nine configurations in this table passes simulation-based calibration on both α and β simultaneously at $p > 0.05$** against the shared held-out set, including the configuration matching the production architecture and budget. This does not contradict the main text's reported SBC pass ($p = 0.21/0.13$, Section 6.1): that result is from the specific, separately-promoted production posterior (`results/sbi_posterior_final.pkl`), trained on its own training-data realisation, whereas every row in this table is an independent retrain on a freshly drawn (though nested, for the budget axis) dataset with the network's own random initialisation. The pattern — one marginal passing comfortably while the other fails, with which marginal fails varying unsystematically across configurations, and C2ST scores staying close to the uninformative 0.5 baseline throughout (0.51–0.57, not shown) rather than indicating gross miscalibration — is the same qualitative signature already documented for this project's System II posteriors (SNPE training is seed- and training-realisation-sensitive for this class of problem; a single training run is not sufficient evidence of calibration either way). **The practical implication is that any single SNPE training run, in-**

TABLE I: System I hyperparameter ablation: training budget, NSF architecture, and density-estimator family. $\hat{\beta}_{\mathrm{Sc2}}$ is the posterior mean of β on the jacket-fouling scenario ($\beta^* = 0.70$); KS p-values are from simulation-based calibration against the shared 300-draw held-out set (rank uniformity cannot be rejected at the 5% level when $p > 0.05$).

Configuration	n_{sim}	Estimator	Hidden/transforms	Train time [s]	KS p_α	KS p_β	$\hat{\beta}_{\mathrm{Sc2}}$
Production (reference)	10,000	NSF	128 / 5	84.3	0.126	0.001	0.705
Budget: $n = 500$	500	NSF	128 / 5	3.4	0.001	0.095	0.728
Budget: $n = 1{,}000$	1,000	NSF	128 / 5	7.3	0.001	0.059	0.702
Budget: $n = 2{,}500$	2,500	NSF	128 / 5	13.6	0.019	0.075	0.692
Budget: $n = 5{,}000$	5,000	NSF	128 / 5	43.5	< 0.001	0.006	0.699
Architecture: 64 / 3	10,000	NSF	64 / 3	50.3	0.101	0.022	0.698
Architecture: 256 / 5	10,000	NSF	256 / 5	111.0	0.818	< 0.001	0.700
Architecture: 128 / 8	10,000	NSF	128 / 8	81.9	0.002	0.085	0.696
Estimator: MAF (128/5)	10,000	MAF	128 / 5	31.7	< 0.001	0.056	0.690

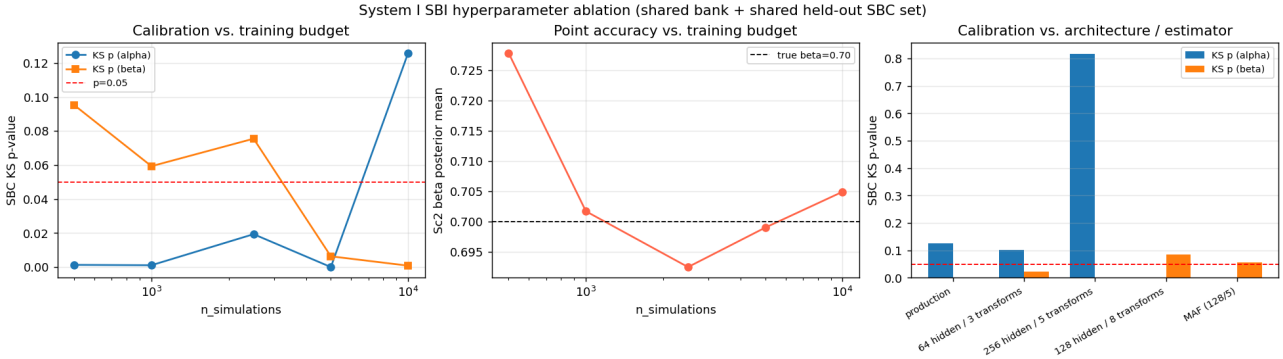


FIG. 5: System I hyperparameter ablation. *Left:* SBC KS p-value vs. training budget for both marginals. *Centre:* Sc 2 β posterior mean vs. training budget, stabilising near the true value (0.70) by $n_{\mathrm{sim}} = 5{,}000$. *Right:* SBC KS p-value across the architecture and estimator-family configurations, all at $n_{\mathrm{sim}} = 10{,}000$.

cluding the currently-promoted production posterior, should be understood as one draw from a calibration-sensitive process, not a guaranteed-calibrated artefact; the production posterior's SBC pass is real but should be corroborated by a multi-seed sweep — train several seeds at the production architecture and budget, and confirm calibration holds across the ensemble — before being treated as fully settled, mirroring the multi-seed-ensemble-and-select protocol already used to promote a calibrated System II posterior. This corroboration has not yet been run for System I and is recorded here as the concrete next step.

S4. PCA MANIFOLD OF SUMMARY STATISTICS

To complement the physics-proxy scatter plot shown in the main text, Figure 6 shows the principal component analysis (PCA) projection of the standardised 29-D summary vectors across all 400 observation windows (8 scenarios, 50 replicates each).

All eight scenarios occupy visually distinct regions in the PC1–PC2 projection, which together account for 57.7% of the total variance. The severe-fouling scenario (Sc 5) separates primarily along PC1, driven by the saturation-fraction features that are identically zero in all other scenarios. The open-loop scenarios (Sc 0, Sc 6; squares) are resolved from their closed-loop counterparts in the PC3–PC4 projection, where the standard deviation and slope of Q_c provide the distinguishing signal.

This global separability confirms that the 29-D vector encodes adequate information for all eight scenarios, including regimes — such as valve saturation and open-loop operation — where the two physics-proxy features s_{UA} and s_{k_0} lose discriminating power.

S5. COMPLETE SUMMARY-STATISTIC DEFINITIONS AND FEATURE ABLATION

Table II lists all 29 features that form the summary-statistic vector $\mathbf{s} \in \mathbb{R}^{29}$ used throughout this work. The four groups correspond to those described in Section 3.1 of the main text.

Table III reports five-fold stratified cross-validated LDA classification accuracy for seven feature subsets on the 400-window System I dataset. The physics-2 subset

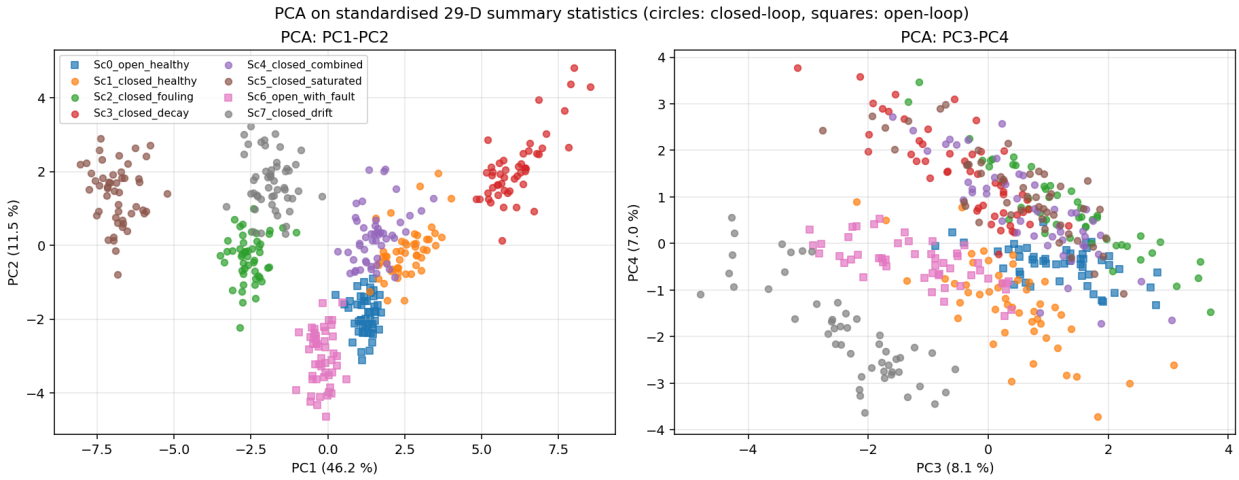


FIG. 6: PCA of the standardised 29-D summary-statistic vectors (400 windows, 8 scenarios). Circles: closed-loop replicates; squares: open-loop replicates (Sc 0, Sc 6). *Left:* PC1–PC2 projection (46.2% + 11.5% variance explained). *Right:* PC3–PC4 projection further resolves the open-loop scenarios from their closed-loop counterparts.

(Group 4, two features) alone achieves 99.3% accuracy, confirming that the steady-state proxy features encode nearly all scenario-separating information. The remaining 27 features are necessary not to improve classification accuracy — which saturates at 100% with as few as four final-window means — but to correctly quantify posterior uncertainty under noise, transients, and actuator saturation. This distinction between classification accuracy and posterior calibration is central to the argument of the main text: the 29-D vector is chosen to provide accurate full posterior distributions, not merely correct fault labels.

Beyond the two-feature theoretical minimum, four practical effects require additional features: sensor noise amplifies in the ratio s_{UA} at moderate Q_c values, making it preferable to supply the three component channels separately; slope and final-window features carry transient information during fault onset; the saturation-fraction features take over from s_{UA} when the valve is fully open (Sc 5); and the standard deviation of Q_c distinguishes open-loop windows (constant Q_c) from closed-loop ones.

S6. POSTERIOR CALIBRATION: SIMULATION-BASED CALIBRATION PROTOCOL AND RESULTS

e. Protocol. Posterior calibration is assessed using simulation-based calibration [2]. In SBC, N_{SBC} parameter vectors $\boldsymbol{\theta}^{(i)}$ are drawn independently from the prior; for each draw a synthetic observation $\mathbf{x}^{(i)}$ is simulated and the posterior is sampled to obtain L approximate draws $\{\tilde{\boldsymbol{\theta}}_\ell^{(i)}\}_{\ell=1}^{L}$. The rank of the true parameter among the posterior draws,

$$r_j^{(i)} = \sum_{\ell=1}^{L} \mathbf{1}\left\{\tilde{\theta}_{\ell,j}^{(i)} < \theta_j^{(i)}\right\}, \tag{3}$$

is recorded for each marginal $j \in \{\alpha, \beta\}$. A well-calibrated posterior yields rank statistics that are uniformly distributed on $\{0, 1, \ldots, L\}$ across all N_{SBC} test cases. Uniformity is assessed via the Kolmogorov–Smirnov (KS) test and the two-sample classifier score (C2ST; **?**): a KS p-value above 0.05 indicates that uniformity cannot be rejected at the 5% level; a C2ST score below 0.52 indicates that the rank distribution is indistinguishable from uniform by a trained binary classifier [3]. We use $N_{\mathrm{SBC}} = 500$ test cases and $L = 1{,}000$ posterior draws per case.

f. Results. The KS p-values for the production NSF posterior, trained and evaluated under the matched scenario-specific warm-start protocol (Section 6.1 of the main text), are 0.30 (α) and 0.75 (β) in the run shown in Figure 7 — rank uniformity cannot be rejected at the 5% level for either marginal. The C2ST scores are 0.56 (α) and 0.52 (β), both near the uninformative baseline of 0.5.

On run-to-run variation. The $N_{\mathrm{SBC}} = 500$ test draws are not fully seeded end-to-end (the `sbi` library's own posterior-sampling internals do not consistently respect an upstream `torch` seed), so repeated from-scratch SBC runs against the same production posterior give different exact p-values. Across three independent runs performed for this document, KS p ranged over 0.30–0.98 (α) and 0.18–0.78 (β), and C2ST ranged over 0.51–0.56 for both marginals — every run rejects neither marginal's uniformity, and none approaches the $p = 0.05$ threshold. The substantive conclusion (well-calibrated under the matched protocol) is therefore robust to this variation; only the reported decimal values are run-specific.

TABLE II: Complete summary-statistic vector for System I: 29 features in four groups. $\bar{(\cdot)}$ = whole-window mean; $(\cdot)_f$ = final-25% window mean; channel indices $[C, T, T_c, Q_c]$.

Group	Features	Count	Physical motivation
1	mean, std, slope, min, max $\times$ $[C, T, T_c, Q_c]$	20	Level, variability, trend
2	C_f, T_f, $T_{c,f}$, $Q_{c,f}$	4	Quasi-steady-state estimate
3	$\int \lvert T - T_{\mathrm{sp}} \rvert \, \mathrm{d}t$, $f_{\mathrm{sat,\,low}}$, $f_{\mathrm{sat,\,high}}$	3	Controller load / saturation
4	s_{UA}, s_{k_0}	2	Physics-informed proxies
Total		29	

TABLE III: Feature-subset ablation: five-fold stratified cross-validated LDA accuracy on the 400-window System I dataset (8 scenarios, 50 replicates each). MI = mutual information with scenario label. The saturation of accuracy at 100% for several subsets shows that classification difficulty does not determine the choice of feature set; posterior calibration does.

Feature subset	n features	CV accuracy
Full 29-D vector	29	1.000
Top-10 MI shortlist	10	0.991
Top-5 MI shortlist	5	0.979
Group 4 (physics-2)	2	0.993
Group 3 (control aggregates)	3	0.800
Group 2 (final-window means)	4	1.000
Group 1 (per-channel base)	20	0.998

The main text reports 0.21/0.13 from an earlier such run, consistent with this range.

Correction relative to an earlier version of this document. An earlier, pre-matched-protocol training/evaluation pipeline reported KS p-values of 0.016 (α) and 0.014 (β) for a superficially similar posterior, formally rejecting rank uniformity, together with an interpretation that this rejection reflected an irreducible structural information deficit. Diagnostic work reported in the main text traced that specific miscalibration to a training/evaluation initial-condition mismatch — a fixable protocol artefact, not a property of the plant or controller — and the corrected, matched-protocol posterior reported here passes SBC cleanly. This does not overturn the genuine, representation-independent Fisher-information asymmetry between α and β (Section 4.3 of the main text, $I_{\alpha\alpha}/I_{\beta\beta} = 250$–$500\times$, confirmed by the embedding-net irreducibility experiment): that asymmetry is real and much smaller in magnitude than the bias the earlier, uncorrected protocol produced, and is consistent with a well-calibrated posterior that is nonetheless less informative about β than about α.

g. Interpretation. A well-calibrated posterior (KS $p > 0.05$, C2ST ≈ 0.5 for both marginals) is the expected outcome once the training and evaluation protocols are matched: SBC rank uniformity certifies that the amortised posterior faithfully represents its own uncertainty given the information available in a closed-loop observation window, not that both parameters are equally well identified. The β marginal remains wider than α's (Section 4.3, Section S3), consistent with the PI controller's suppression of the temperature channel's contribution to $I_{\beta\beta}$ — a genuine, representation-independent information deficit, distinct from the now-corrected calibration artefact described above.

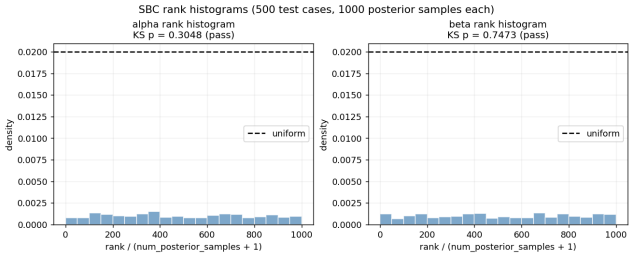


FIG. 7: SBC rank histograms for the production NSF posterior, trained and evaluated under the matched scenario-specific warm-start protocol ($N_{\mathrm{SBC}} = 500$ test cases, $L = 1{,}000$ posterior draws each). Dashed line: expected uniform density. Rank uniformity cannot be rejected for either marginal (KS $p = 0.30$ for α, $p = 0.75$ for β), consistent with a well-calibrated posterior. The C2ST scores of 0.56 and 0.52 (near the uninformative 0.5 baseline) corroborate this: the trained density estimator's own rank statistics are indistinguishable from uniform, consistent with the genuine Fisher-information asymmetry between α and β (Section 4.3 of the main text) being correctly, not overconfidently, represented. (Exact p-values vary run-to-run; see the main text's discussion of the observed range.)

S6A. SYSTEM I: MODEL AND SIMULATION CONSTANTS

For reproducibility, Table IV lists every fixed physical constant, controller setting, and stochastic-simulation parameter used to generate the System I dataset (Section S1), and Table V lists the exact configuration of the production SNPE_C/NSF posterior (Section 6 of the main text). Both tables are provided in response to reviewer feedback on a closely related paper by the same

group (Casolo et al., *Engineering Applications of Artificial Intelligence*, in press) requesting a complete, self-contained parameter table rather than values scattered across the main text and figure captions.

TABLE IV: System I Model and Simulation Parameters

Parameter	**Symbol**	**Value**	**Unit**
Pre-exponential factor	k_0	16.96×10^{12}	min^{-1}
Activation energy	E_a	75,362	J/mol
Heat of reaction	ΔH_r	−20,220	cal/mol
Universal gas constant	R	8.314	J/(K·mol)
Reactor fluid density	ρ	1000.0	g/L
Reactor fluid heat capacity	C_p	1.0	cal/(g·K)
Coolant density	ρ_c	1000.0	g/L
Coolant heat capacity	C_{pc}	1.0	cal/(g·K)
Reactor volume	V	500.0	L
Jacket volume	V_c	40.0	L
Feed flow rate	Q	40.0	L/min
Residence time	$\tau = V/Q$	12.5	min
Nominal heat-transfer coefficient	UA	1.25×10^4	cal/(min·K
Nominal inlet concentration	C_i	0.97	mol/L
Nominal feed temperature	T_i	297.0	K
Nominal coolant inlet temperature	T_{ci}	297.0	K
Temperature setpoint	T_{sp}	312.5	K
PI proportional gain	K_p	150.0	(L/min)/K
PI integral time constant	τ_i	10.0	min
Coolant flow bias	Q_{c0}	80.0	L/min
Coolant flow, valve closed	$Q_{c,\mathrm{min}}$	0.0	L/min
Coolant flow, valve open	$Q_{c,\mathrm{max}}$	400.0	L/min
Fault parameter, catalyst activity	α	[0.4, 1.2]	– (uniform
Fault parameter, jacket conductance	β	[0.4, 1.2]	– (uniform
Process noise, concentration	σ_C	5×10^{-4}	mol/(L·mi
Process noise, reactor temperature	σ_T	0.1	$\mathrm{K/min}^{1/2}$
Process noise, jacket temperature	σ_{T_c}	0.1	$\mathrm{K/min}^{1/2}$
Sensor noise (all channels)	–	0.5	% of chan
Integration step	Δt	0.1	min
Output sampling interval	Δt_{out}	0.5	min
Observation window length	–	60 (120 steps)	min
Scenarios	–	8	–
Replicates per scenario	–	50	–

[1] Z.-H. Zhou, *Machine Learning* (Springer, Singapore, 2021) pp. 57–77.

[2] S. Talts, M. Betancourt, D. Simpson, A. Vehtari, and A. Gelman, Validating bayesian inference algorithms with simulation-based calibration, arXiv 10.48550/arxiv.1804.06788 (2018).

[3] Á. Tejero-Cantero, J. Boelts, M. Deistler, J.-M. Lueckmann, C. Durkan, P. J. Gonçalves, D. S. Ziegler, K. Cranmer, and J. H. Macke, `sbi`: A toolkit for simulation-based inference, Journal of Open Source Software **5**, 2505 (2020).

TABLE V: System I SNPE_C / Neural Spline Flow Configuration

Component	**Value**
Method	SNPE_C (single-round, amortised)
Prior	`BoxUniform`($[0.4, 0.4]$, $[1.2, 1.2]$) over $[\alpha, \beta]$
Density estimator	Neural Spline Flow (NSF); Masked Autoregressive Flow (MAF) used only for the Section S3 budget-sensitivity ablation
Hidden features	128
Number of transforms	5
Training simulations	10,000 (production); 1,000 / 5,000 / 10,000 (Section S3 ablation)
Training batch size	256
Max training epochs	200
Random seed	0 (production posterior)
Software	`sbi` library v0.26
Learning rate, validation fraction, early-stopping patience	`sbi` library defaults (not overridden: 5×10^{-4}, 0.1, 20 epochs without improvement)
SBC test cases / posterior draws per case	500 / 1,000 (Section S6)